Agenda

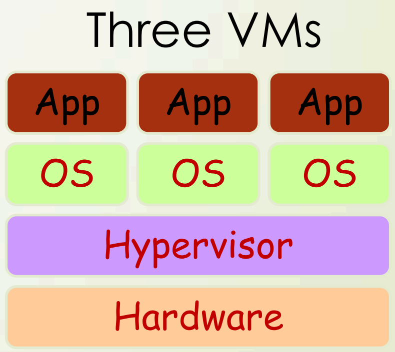
- What is Cloud
- Cloud Service Models
- Deployments
- Advantages
- Challenges and Potential Solutions

Four Deployment Models

- **Public cloud** (off-site and remote): resources are dynamically provisioned on an on-demand, self-service basis over the Internet, via web applications/web services, open API, from a third-party provider who bills on a utility computing basis.
- **Private cloud**: corporations discover the benefits of consolidating shared services on virtualized hardware deployed from a primary datacenter to serve local and remote users
- **Hybrid cloud**: some computing resources are on-site (on premise) and some are off-site (public cloud). To leverage cloud solutions that are too costly to maintain on-premise, like disaster recovery, backups and test environments
- **Community cloud**: several organizations/corporations with similar requirements share common infrastructure. Somewhere between public and private cloud

Virtualization

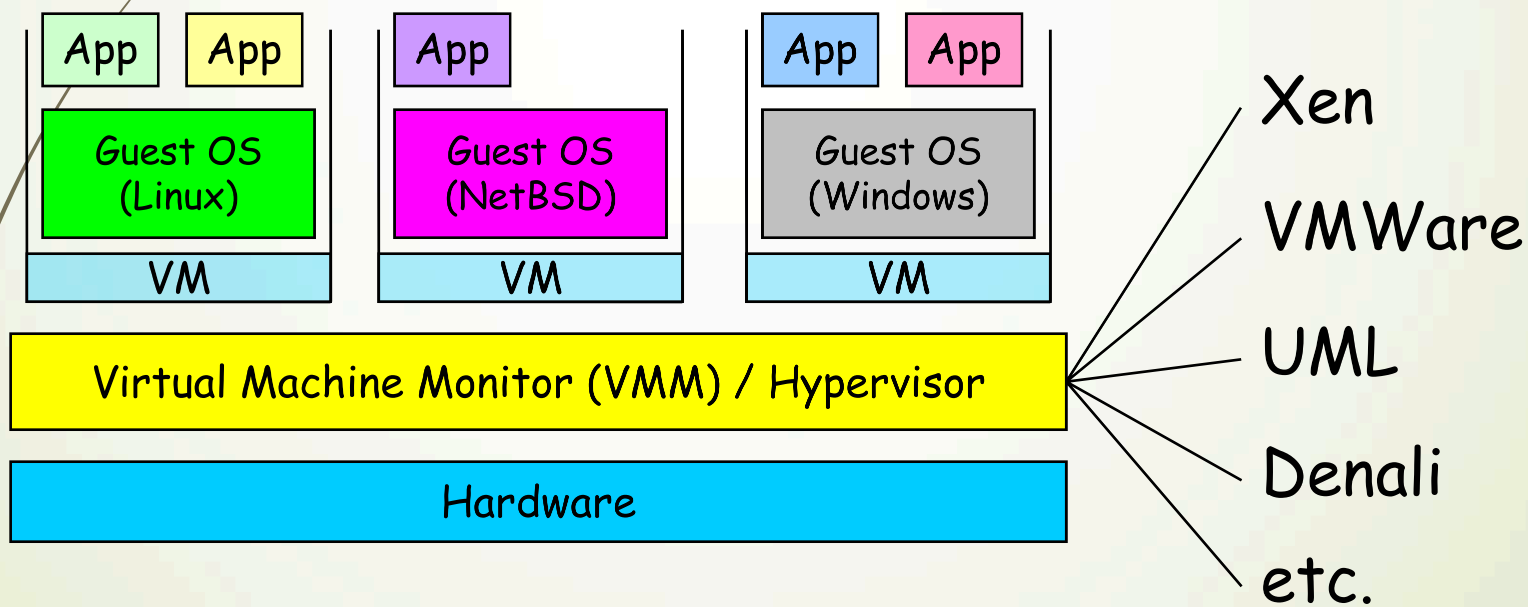
- Virtual workspaces
  - An execution environment that can be made dynamically available
  - Resource quota (e.g. CPU, memory)
  - Software configuration (e.g. O/S, provided services)
- Implement on Virtual Machines (VMs):
  - Abstraction of a physical host machine (a.k.a. bare metal)
  - Hypervisor intercepts and emulates instructions from VMs, and allows management of VMs,
  - VMWare, VirtualBox, Xen, etc.
- Provide infrastructure API:
  - Plug-ins to hardware/support structures



Virtualized Stack

Virtual Machines

- multiple virtual machines on a physical machine
  - emulation
  - full virtualization
  - para-virtualization



Para-virtualization (e.g., Xen) is very close to raw physical performance!

Benefits of Virtualization

- Easier to create new machines, backup machines, etc., ← running environment, e.g., for our programming asignments
- Easier migration
- more machines than the physical ones ← running experiments
- Timeshare lightly loaded systems on one host
- suspend and resume VMs ← debugging and troubleshooting
- Run operating systems where the physical hardware is unavailable ← emulation
- Run legacy systems ← Mupen64Plus (N64), DeSmuME (DS)

Sample IaaS Service

Load image to S3

Boot your image

Open up ports

SSH into your image

Run your binaries

- Amazon Elastic Compute Cloud (EC2):
  - Elastic, marshal 1 to 100+ PCs
  - Machine Specs to meet your needs
  - Fairly cheap, if your jobs are not resource hungry
- Powered by Xen
  - Different from VMware as uses "para-virtualization" where the guest OS is modified to use special hyper-calls
  - Hardware accelerated virtualization by Intel (VT-x/Vanderpool) and AMD (AMD-V)
  - Supports "Live Migration" of a virtual machine between hosts.
- Linux, Windows, and OpenSolaris
- Management Console

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Advantage of Cloud Computing (1/2)

- Lower computer costs
  - No more dedicated powerful and expensive workstations that are seldom used
  - Desktop PCs can be lighter-weight since software runs in the cloud
- Improved performance
  - Can request for more powerful cloud server on-demand
  - Your PCs may run faster due to lower load
- Universal availability (to, but not limited to, documents)
  - Any client device will work
- Higher reliability
  - Both data and hardware

Advantage of Cloud Computing (2/2)

- Reduced software costs
  - Most cloud applications, like Google Doc Suite, is free ← **But is it really free?**
  - Some commercial software may be installed on a shared cloud server
- No more foreground software update if it runs in the cloud
- Large storage space
  - Some cloud storage service providers offer (virtually) unlimited storage space
  - Cloud storage automatically cache files on your local harddisks
  - Implicit version control
  - Easier for collaborations over the Internet

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Need a Constant Internet Access

- Limitation: Pure cloud computing requires always-on Internet access
- Solution approach: keep temporal states on clients
  - cloud storage systems (like dropbox) sync with the cloud server whenever Internet access is available
  - Waze leverage cached maps to provide directions even when the Internet is temporarily out
- Solution approach: edge clouds or cloudlets may be leveraged when Internet connectivity is unavailable
  - Even more general, an dynamically deployed VMs to computing devices in proximality
- Typically need a middleware-based solution, instead of reinventing the wheel all the time
- Other thoughts?

Can Be Slow

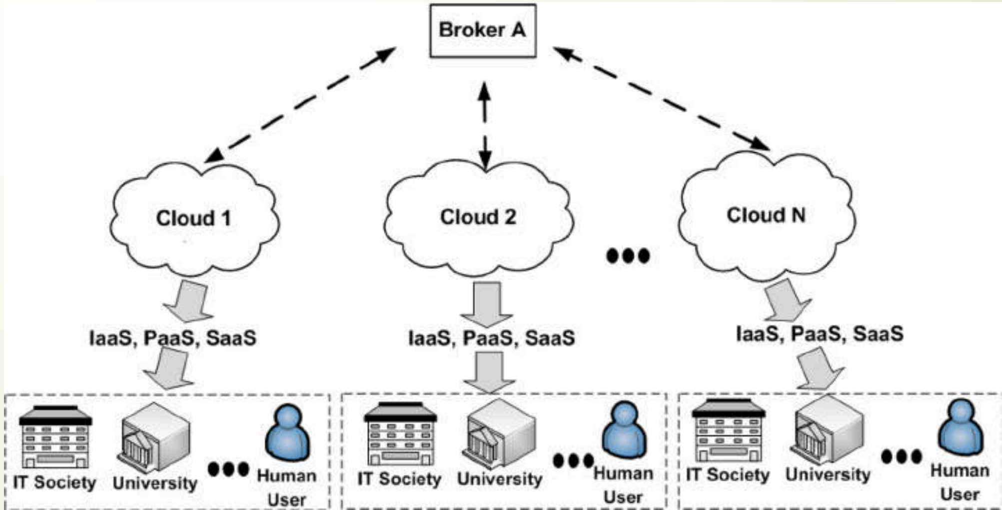
- Limitation: Too many factors may cause slow responsiveness of cloud applications
  - Insufficient reserved resources, busy cloud server (e.g., when backing up the data), long network latency, and insufficient network bandwidth ← determining the root cause itself is a research problem
- Solution approach: Through modeling (cloud) application resource consumption under different quality target, we may better choose the optimal resource levels to reserve
- Solution approach: Find closer data center server or (better) the data center with less-congested networks
- Solution approach: Compress data (lossless or lossy) or feature extraction before transmitting data
- Any other thoughts?

Data Security

- Limitation: Once the data are stored in the cloud, unauthorized users may gain access to your confidential data
- Solution approach: Save only parts of data on a cloud server provider
- Solution approach: Encrypt/decrypt your data on your PC ← but then the PC workload is increased
- Solution approach: Variants of partial encryption offloading to an edge or cloud servers have also been proposed
- Even with encrypted data, many inference can and have been done based on traffic statistics (without looking into the payload)
- Other thoughts?

Segmented Cloud Service Providers

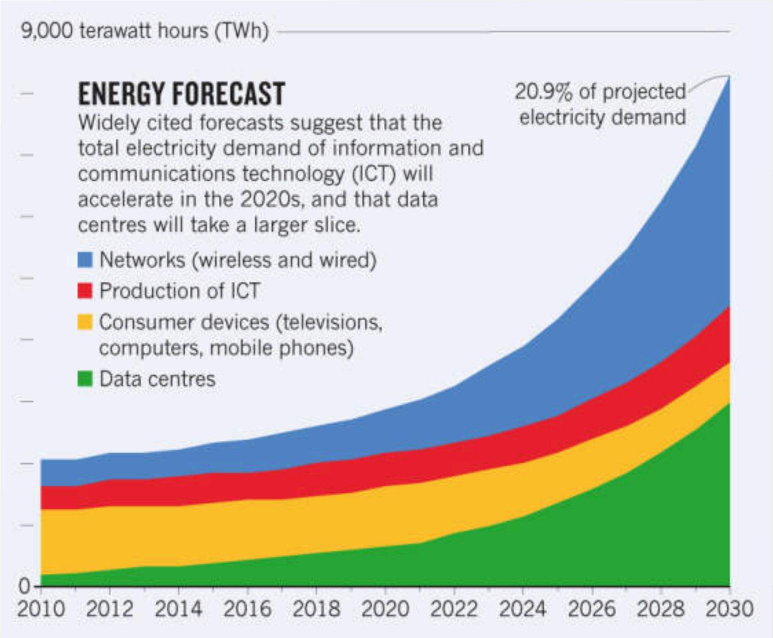
- Limitation: Each cloud systems uses different protocols and different APIs → may not be possible to run the same application on different cloud based systems
- Solution approach: Open-source projects, such as Open Stack, could be used to build private cloud
  - Then migrate to public clouds
- Solution approach: Broker based federated clouds
- Other thoughts?



Source: A Decentralized Approach For The Management Of Federated Cloud Environments

Energy Consumption

- Limitation: Data centers become the major source of high electricity demands ← Cooling is an issue if we put too many computers together
- Solution approach: Better data center design to cool the computers down← several fun papers from Google to read
- Solution approach: Better job scheduling? For example, more balanced workload would reduce the extreme temperature at some data centers
- Solution approach: Crowdsourcing computing, storage, and networking resources of otherwise idling PCs to reduce the loads on data centers ← would it be really more energy efficient?
- Other thoughts?



[Source] Andrae, A. & Edler, "*On Global Electricity Usage of Communication Technology*: Trends to 2030 " T. Challenges 6, 117–157 (2015)

Questions

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